

P²-ViT: Power-of-Two Post-Training Quantization and Acceleration for Fully Quantized Vision Transformer

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Abstract—Vision Transformers (ViTs) have excelled in computer vision tasks but are memory-consuming and computation-intensive, challenging their deployment on resource-constrained devices. To tackle this limitation, prior works have explored ViT-tailored quantization algorithms but retained floating-point scaling factors, which yield non-negligible re-quantization overhead, limiting ViTs’ hardware efficiency and motivating more hardware-friendly solutions. To this end, we propose *P²-ViT*, the first Power-of-Two (PoT) post-training quantization and acceleration framework to accelerate fully quantized ViTs. Specifically, as for quantization, we explore a dedicated quantization scheme to effectively quantize ViTs with PoT scaling factors, thus minimizing the re-quantization overhead. Furthermore, we propose coarse-to-fine automatic mixed-precision quantization to enable better accuracy-efficiency trade-offs. In terms of hardware, we develop a dedicated chunk-based accelerator featuring multiple tailored sub-processors to individually handle ViTs’ different types of operations, alleviating reconfigurable overhead. Additionally, we design a tailored row-stationary dataflow to seize the pipeline processing opportunity introduced by our PoT scaling factors, thereby enhancing throughput. Extensive experiments consistently validate *P²-ViT*’s effectiveness. Particularly, we offer comparable or even superior quantization performance with PoT scaling factors when compared to the counterpart with floating-point scaling factors. Besides, we achieve up to 10.1× speedup and 36.8× energy saving over GPU’s Turing Tensor Cores, and up to 1.84× higher computation utilization efficiency against SOTA quantization-based ViT accelerators. Codes are available at <https://github.com/shihuihong214/P2-ViT>.

Index Terms—Power-of-two, post-training quantization, Vision Transformer, ViT accelerator, fully quantized ViT.

I. INTRODUCTION

THANKS to the powerful capability of Transformers’ self-attention mechanism in extracting global information, Vision Transformers (ViTs) have shown great potential and achieved remarkable success in various computer vision (CV) tasks [1]–[3]. Despite their promising performance, ViTs typically have more parameters and intensive computations than their convolution-based counterparts, yielding higher memory footprint and energy costs during inference. For example, ViT-Large [1] involves 307M parameters and 190.7G FLOPs for

achieving 87.76% top-1 accuracy on ImageNet [4]. This challenges the deployment of ViTs on resource-constrained edge devices, calling for effective model compression solutions.

Model quantization, which converts weights and/or activations from floating-point ones to low-precision integers without modifying network architectures, stands out as a generic and effective model compression technology. Thus, to facilitate ViTs’ deployment, it is natural to adopt quantization to reduce both their memory usage and computational costs. Specifically, [5] and NoisyQuant [6] consider ViTs’ specific algorithmic properties to quantize ViTs’ linear operations (i.e., matrix multiplications); FQ-ViT [7] introduces several tailored techniques to quantize ViTs’ non-linear operations (i.e., LayerNorm and Softmax), thus offering fully quantized ViTs. However, despite the effectiveness of existing ViT quantization methods, they generally target quantization of linear [5], [6], [8]/non-linear [7], [9]) operations while overlooking and retraining scaling factors in floating-point. This yields non-negligible re-quantization overheads and hinders ViTs’ integer-only inference (see Fig. 2a), limiting their speedup on existing hardware platforms [10], [11] and motivating the exploration for more hardware-efficient solutions.

In parallel, prior works [9], [12]–[16] have developed dedicated accelerators to boost ViTs’ hardware efficiency from the hardware perspective. For example, ViTCoD [12] first leverages pruning to reduce ViTs’ redundancy, then develops a dedicated accelerator to gain hardware speedup. Additionally, VAQF [14] and Auto-ViT-Acc [15] implement the acceleration of ViTs on FPGAs with FPGA-aware automatic quantization. While dedicated accelerators can significantly enhance hardware efficiency, existing ViT accelerators [14]–[16] typically focus on the acceleration of matrix multiplications while ignoring the remaining non-linear operations and energy-consuming re-quantization process involved in ViTs. These oversights hinder the acceleration potential through techniques like layer fusion and pipeline processing [17], thus limiting ViTs’ achievable hardware efficiency and real-world applications. Consequently, it is desirable to build dedicated accelerators to overcome these limitations, i.e., to accelerate fully quantized ViTs alongside minimizing re-quantization overhead to embrace the layer fusion opportunity, which yet is still under-explored.

To this end, to facilitate real-world deployment of ViTs with maintained accuracy, we make the following contributions:

- We propose **P²-ViT** (see Fig. 1), a **Power-of-Two (PoT)** post-training quantization and acceleration framework to

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